

Sunidhi

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EDUCATION

Duke University (CGPA: 3.7/4.0)

Master of Engineering Biomedical Engineering- Medical Devices

Courses: ADM (Advanced Manufacturing and Design), Design Health, Cardiac Electrophysiology, Quality Systems, Biomaterials, Signals

May 2026

Durham, NC

Manipal University Jaipur (CGPA: 4.0/4.0)

Master of Science in Biotechnology-Industrial Engineering

Courses: Physiology, Biostatistics, Genetics, Material Design, Industrial Manufacturing, Biosensor design, Medical Instrumentation

May 2023

Jaipur, IND

EXPERIENCE

Johnson & Johnson Medtech (Electrophysiology)

April 2025- December 2025

NPD Quality Engineer Co-op

Irvine, CA

- Automated PFA cycle testing by designing and prototyping an ablation actuator integrated with the TruePulse generator to run 100 cycles per cable, eliminating manual testing and saving **\$38K**.
- Led randomized, blinded Design Verification Testing (DVT) across 30 commercial catheters with 3 operators and 3 attributes (**270 total trials**), achieving **100%** results and supporting release under design controls.
- Authored a new standard test method (**STM**) for ablation performance testing and Luer-hub pull test, improving precision **18%** and repeatability **22%** across commercial catheter builds.
- Enabled root-cause-driven design updates by building an arcing visualization tool that detected micro-discharge events, reducing arcing incidents **64%** during verification testing and improving visualization **100%**.
- Reduced PU application variability by running **DOE** and updating **SOP** controls based on findings, lowering process variability **20%** and improving manufacturing consistency.
- Led root cause analysis using a fishbone analysis for a recurring cleanroom deviation, initiated CAPA, and deployed a Power BI dashboard to trend nonconformances, reducing repeat deviations 20% in 3 months.

Duke Stimulation Centre

October 2024- April 2025

Clinical R&D Engineer

Durham, NC

- Conducted **40-plus hours** of shadowing and **12 physician interviews**, translated clinical findings into 18 traceable design inputs and 32 measurable engineering specifications for a next-generation electrical cystoscope.
- Designed and built an ergonomic electrical cystoscope prototype within a \$100 budget, converting design inputs into manufacturable architecture that reduced setup time by **20 minutes** per procedure.
- Performed **DFMEA** and usability risk assessments (**uFMEA**) across 15 failure modes, implementing risk controls and DFM improvements that reduced user-related defects **15%** and functional defects by **21%**.
- Translated 8 procedural steps into standardized checklists and failure-response guides, reducing use errors and improving usability consistency across 10 simulated-use sessions.
- Reduced repeat high severity use errors **by 37%** after implementing design mitigations identified through analysis of **24 logged** use events and prepared human factors validation documentation aligned with **IEC 62366** and FDA HF guidance.

S&M Industries

May 2022-July 2024

Manufacturing and Operations Head

India

- Led **ISO 13485**-controlled Class I cleanroom manufacturing for 4 SKUs across 2 production lines, overseeing end-to-end operations (documentation, IPC, release, and dispatch) to deliver ~10,000 units/month with **98%** OTIF and **>97%** FPY.
- Drove defect containment and disposition for line events, executed bracketing, initiated CAPA investigations, and implemented corrective actions with QC and Ops, reducing defect rate **35%** and scrap and rework **20%** in one quarter.
- Established SPC-based process controls and capability monitoring, deployed fixtures and manufacturing aids, reducing cycle time **18%** and increasing equipment uptime **12%**.
- Designed and executed Operational Qualification (**OQ**) studies to establish critical process parameter ranges, performing **DOE** and statistical analysis that reduced process variability **18%**.

PROJECT

Spinal Biopsy and Aspiration Needle

November 2024-December 2024

- Conducted root-cause and user-needs analysis through clinician interviews and procedural mapping, identifying ergonomic and torque-control deficiencies in existing biopsy systems to guide design inputs.
- Engineered a dual-shaft, torque-balanced prototype in SolidWorks, performing tolerance stack-ups and finite-element simulations to validate mechanical integrity, load response, and manufacturability under **ISO 13485** controls.

Sleep-Apnea

September 2024- April 2025

- Conducted over **70 hours** of clinical observation and patient interviews to identify key usability and comfort challenges in existing CPAP therapy, translating unmet clinical needs into actionable design inputs for a next-generation tongue stabilization device.
- Applied structured design control and risk management (DFMEA, design reviews) to evaluate design safety, biocompatibility, and manufacturability, ensuring traceable alignment with **FDA 510(k)** pathways, **ISO 13485** quality systems, and **ISO 14971** risk standards.
- Designed, fabricated, and validated a functional prototype using silicone and carbon-fiber composites, and testing demonstrated a 20% improvement in patient comfort and consistent airway patency under simulated clinical conditions.
- Logged and categorized **24 use errors** across formative evaluations, prioritizing **9 high-severity** events and implementing targeted design controls that mitigated repeat occurrence before validation.

SKILLS

- Design and Stimulation** SolidWorks, AutoCAD, Fusion360, 3D-Slicer (Medical Image Reconstruction), Arduino, Electrophysiology
- Manufacturing** IQ, 3D-Printing, SPC, DFM, LBM, Design Control Framework, Clinical roadmaps, Risk assessment
- Data Analysis** Six Sigma, Regression Analysis, SQL, Pandas, NumPy, Minitab, Gage R&R, Verification & Validation
- Quality & Regulatory** ISO 13485, FDA 510(k), CAPA, Gage R&R, QMS (21 CFR Part 820), IEC 60601-1-6 / IEC 62366-1, IEC 60601-1